Lecture 9: Introduction to Classification

Classification

- So far, we've mostly talked about regression – predicting a continuous value given some inputs & training data

- Many examples of machine learning (some we've mentioned) are based on classification rather than continuous data

Classification

- Want to learn f: X -> Y
  - X is the feature(s) that you will use to predict
  - Y is the target classes

- Conditional probability: p(Y|X)
  - That is, you want to know the probability that you belong to a particular class given the features of the data

Remember: Bayes theorem

- $p(A|B) = [p(B|A) * p(A)] / p(B)$

  For our purposes, p(B), the probability of our data without any other information is constant, so:

- $p(A|B) \alpha p(B|A) * p(A)$

  This is useful to us because we don't care about the exact probability, we just want to optimize our model

"What is the most probable hypothesis given our training data?"

Bayes optimal classifier

- Also called:
  - Bayes optimal learner
  - Bayes classifier
  - Bayes optimal decision boundary
  - Bayes optimal discriminant function

- The specific question answered by the Bayes optimal classifier is *“what is the most probable classification of the new instance given the training data?”*

Bayes optimal classifier

- We want to calculate the probability of a new instance (v_j) given the training data (D), given the space of hypotheses (H):

  $$P(vj \mid D) = \text{sum } \{h \text{ in } H\} P(vj \mid hi) * P(hi \mid D)$$

- Selecting the outcome with the highest probability is Bayes *optimal* classification:

  $$\max \text{ sum } \{h \text{ in } H\} P(vj \mid hi) * P(hi \mid D)$$

Bayes optimal classifier

Any system that classifies new instances according to [the equation] is called a Bayes optimal classifier, or Bayes optimal learner. No other classification method using the same hypothesis space and same prior knowledge can outperform this method on average.

-Machine Learning, 1997. Page 175

Bayes optimal classifier

- Due to uncertainty in the training set and inaccuracies in the hypothesis space, the model will make errors
  - However, this error is the minimum possible error
  - **Bayes error**: the minimum possible error that can be made while making predictions

- Note that this methodology can be applied to regression as well as classification (although we won't)

Bayes optimal classifier

Note that it is a theoretical model. A gold standard, an unattainable ideal to pursue, rather than a method that we can use. This is because we cannot know the conditional probability of Y given X.

Bayes optimal classifier

- It is not computationally feasible, or statistically possible, to implement a proper Bayes optimal classifier, so we use simplifications. Two common simplifications are:
  - **Gibbs Algorithm**. Randomly sample hypotheses biased on their posterior probabilities
  - **Naïve Bayes**. Assume that variables in the input are conditionally independent
  - Really, many non-linear machine learning algorithms that are able to make predictions are close approximations of Bayes optimal classifier in practice

Back to where we started: classification

- The problem of classification is to assign a label to a given input data sample

- Bayes theorem is an obvious choice to deal with this, but has severe limitations in most applications:
  - Requires *a lot* of training data
  - Is computationally very expensive
  - This is because of how the states of your training (and test) data relate to one another
    - If you want to predict that an email is spam, the probabilities that it contains the word “congratulations” and is from a previously uncontacted address are related to one another.

Naïve Bayes

- But what if the states *did not* depend on one another?
  - Using the word “congratulations” has no impact on whether you’ve contacted that sender in the past (and vice versa)
  - Fidgeting in an airport has no impact on whether or not you look away from security personnel (and vice versa)
  - …
- Naïve Bayes makes this assumption – the assumption of independence

Naïve Bayes – slight math

- p(you belong to this class|thing you learn from 1 and thing you learn from 2 and thing you learn from 3…thing you learn from n) = something very hard to calculate
  - You need *TONS* of data
  - It'll take forever and half to compute

Unless…

- All of the things you learn from are independent!
- Then, p(you belong to this class|all that stuff) = p(you belong to this class|thing 1) * p(you belong to this class|thing 2) * …

Naïve Bayes – slight math example

- p(email is spam|it is from unknown sender & it uses non-letter text & it isn't from @conncoll.edu…& it says "you won") = something very hard to calculate
  - You need *TONS* of data
  - It'll take forever and half to compute

Unless…

- All of the things you learn from are independent!
- Then, p(email is spam|all that stuff) = p(email is spam|unknown sender) * p(email is spam|uses non-letter text) * …

Naïve Bayes

- p(you belong to this class|all that stuff) = p(you belong to this class|thing 1) * p(you belong to this class|thing 2) * …
  - p(you belong to this class|thing 1) α p(thing 1|you belong to this class) * p(you belong to this class)
    - p(you belong to this class) is trivial to estimate: examples of this class in training data / total examples in your training data

      -or-

    - p(spam email) = # spam email in training data / total email in training data

Naïve Bayes

- p(you belong to this class|thing 1) α p(thing 1|you belong to this class) * p(you belong to this class)
  - p(thing 1|you belong to this class) isn't very hard either (although harder)
    - p(unknown sender|it's a spam email)
    - It depends very strongly on what kind of data you're looking at
      - Is it numerical? Binary? Categorical?
      - You can envision using temperature as a feature that you learn on. Or sunny vs not. Or season.
      - You may (and often will) have different kinds of data in your algorithm. This isn't a problem, just use the appropriate method for each feature!
      - Binary data uses the binomial distribution, categorical data uses multinomial distribution, and numeric data uses Gaussian distribution
      - You can also just approximate from the data.